

ChatDesk Development Phase

Development Timeline: July 6 – August 6, 2026

Roles:

Jefferson: Lead Full-Stack / Backend Developer

Veejay: Full-Stack / Backend Developer

Keith: Frontend Developer (React Native Mobile)

Zaki: Frontend Developer (ReactJS Web)

Sprint 0: Planning and Setup

July 6 – July 7

Goal

Prepare the local environments, folder layouts, system contracts, and repository structure.

Tasks

All Developers

- Finalize ChatDesk MVP scope and requirements.
- Establish and clone the Git repository with the main and develop structure.
- Agree on coding standards and variable casing restrictions (snake_case backend, camelCase frontend).

Full-Stack / Backend Developers (Jefferson & Veejay)

- Scaffold the Django REST Framework root project hierarchy.
- Configure local PostgreSQL instances.
- Publish the shared API_CONTRACT.md and ENV_GUIDE.md files locally inside the /docs directory.

Frontend Developers (Keith & Zaki)

- Set up the React Native project template using Expo.
- Set up the ReactJS web dashboard using Vite and Tailwind CSS.
- Prepare empty directory structures for assets and components matching mock endpoints.

Output

- Repository branches structured (main, develop).
- /docs repository folder finalized inside VS Code.
- Baseline structural scaffolding running smoothly on all developer machines.

Sprint 1: Authentication, Core Schemas, and UI Foundations

July 8 – July 15

Goal

Build user identification security protocols, database schemas, and establish the visual layouts for both web and mobile screens.

Backend Tasks

Jefferson (Full-Stack Lead)

- Build the custom tbl_user table models inside Django.
- Configure explicit role designations for student and admin privileges.
- Establish database constraints for user indexing.

Veejay (Backend)

- Develop the registration and login endpoint architectures (/api/v1/auth/register/, /api/v1/auth/login/).
- Integrate secure JWT Token generation, expiration handling, and token blacklisting configurations.
- Implement backend route protection middleware.

Frontend Tasks

Keith (Mobile - React Native)

- Develop the Student login interface layout.
- Develop the Student registration screen.
- Design the skeleton shell of the main mobile conversational container page.

Zaki (Web - ReactJS)

- Build the OSA Admin login validation screen view.
- Scaffold the foundational responsive web dashboard navigation and app grid layout.
- Create table placeholders for ticket management visual modules.

Integration Work

- Connect both frontends to the login validation APIs.
- Validate secure local management of JWT access and refresh tokens.

Sprint 1 Output

- Authentication pipeline fully functional.
- Role-based interface constraints structurally validated.
- Base screens established across both presentation clients.

Sprint 2: Core Intelligent Agent & Ticket Automation Engine

July 16 – July 26

Goal

Implement automated natural language request pipelines via OpenAI integration alongside fallback administrative tracking mechanics.

Backend Tasks

Jefferson (Full-Stack Lead)

- Build the core text extraction and matching logic linking inbound student payloads to tbl_faqs.
- Integrate backend bindings linking to the external OpenAI API endpoint.

- Implement context isolation and strict prompt engineering directives prioritizing Taglish classification capabilities.

Veejay (Backend)

- Construct the schemas and endpoints tracking data records inside `tbl_inquiry_logs`.
- Build the automatic structural logic generating support requests inside `tbl_ticket` upon detecting `unresolved_complex_query` intents.
- Develop structural admin update views for managing ticket status parameters.

Frontend Tasks

Keith (Mobile - React Native)

- Program full asynchronous state handlers inside the student dialogue feed.
- Implement visual presentation formatting differences separating bot responses from human dialogue blocks.
- Add a local historical portal page allowing students to view their ongoing ticket status parameters.

Zaki (Web - ReactJS)

- Build the Admin FAQ Data Grid manager permitting direct insertion, alteration, and removal of structural data records.
- Construct the administrative Ticket Inbox grid view displaying interactive listings of outstanding student requests.
- Implement visual operational status badges separating unassigned, active, and resolved issue rows.

Integration Work

- Perform end-to-end integration mapping active chat dialogue fields through to administrative ticketing updates.
- Verify strict 1-to-1 backend data table constraints link newly instantiated tickets to historical dialogue roots.

Sprint 2 Output

- Automated AI customer query pipelines running stably.
- Dynamic context management working reliably in Taglish.
- Ticketing escalation routing functioning seamlessly upon operational fallback triggers.

Sprint 3: Reporting Diagnostics, System Evaluation, and Cloud Launch

July 27 – August 6

Goal

Finalize quantitative metric dashboards, build data capture linkages, deploy operational infrastructure, and complete quality assurance.

Backend Tasks

Jefferson (Full-Stack Lead)

- Configure the production orchestration templates targeting DigitalOcean server boundaries.
- Isolate final system environment strings within the server dashboard configuration profiles.
- Conduct structural stress assessments mapping payload tracking constraints across concurrent sessions.

Veejay (Backend)

- Develop analytical aggregation routines parsing transaction records within the inquiry history logs.
- Build structured statistical extraction routes outputting total usage volumes, intent frequencies, and open ticket resolution rates.

Frontend Tasks

Keith (Mobile - React Native)

- Program a navigation gateway link referencing the system evaluation questionnaire hosted externally via Google Forms.
- Execute visual polish loops aligning padding, sizing, typographic weights, and asset assets directly alongside the Figma layout mockups.

Zaki (Web - ReactJS)

- Design and render visual analytics charts mapping computational intent densities, traffic frequencies, and service times.
- Build filtering modules allowing admins to isolate logs by category, dates, or individual identification keys.

Integration Work

- Migrate production configurations securely across to live target domains on DigitalOcean.
- Populate database fields with realistic contextual testing scenarios to perform complete walkthrough evaluations.

Sprint 3 Output

- ChatDesk operational platform deployed.
- Diagnostic dashboard accurately rendering aggregated usage data.
- Google Forms evaluation pipeline integration working cleanly on mobile clients.
- Project finalized and ready for the technical presentation panels.